

Computer and Network Security

Tutorial Worksheet B-01

Description. This tutorial worksheet covers the following topics:

- Modular arithmetic
- Affine cipher
- Caesar cipher
- Hill cipher

Exercise 1 (Modular Arithmetic)

1. Compute the following congruence:

- $139 \equiv ? [26]$
- $749 \equiv ? [26]$
- $255 \equiv ? [26]$
- $422 \equiv ? [26]$
- $111 \equiv ? [26]$
- $649 \equiv ? [26]$

2. Compute the following exponentiation:

- $23^3 \equiv ? [30]$
- $11^7 \equiv ? [13]$
- $3^{1000} \equiv ? [23]$
- $31^{500} \equiv ? [30]$
- $88^7 \equiv ? [187]$
- $3^{547} \equiv ? [825]$
- $242^{329} \equiv ? [243]$
- $3^{100} \equiv ? [29]$
- $123^{562} \equiv ? [100]$

3. Compute the following modular multiplicative inverses:

- $3 \cdot 3^{-1} \equiv 1 [26]$
- $3 \cdot 3^{-1} \equiv 1 [43]$
- $3 \cdot 3^{-1} \equiv 1 [57]$
- $7 \cdot 7^{-1} \equiv 1 [26]$
- $7 \cdot 7^{-1} \equiv 1 [43]$
- $7 \cdot 7^{-1} \equiv 1 [57]$

Exercise 2 (Caesar Cipher)

1. Use Caesar cipher to encrypt the following plaintext using the key $K=10$:

“I WANT TO MEET YOU AT DOWN”

2. Use Caesar cipher to decrypt the following two ciphertexts using the key $K=19$:

“PAXG MAX VTM BL TPTR, MAX FBVX PBEE IETR”

“IXHIEX’L WXFHVKTMBV KXINUEBV HY TEZXKBT”

3. Apply brute-force attack pattern to crack the key K that was used to encrypt a plaintext (having the meaning of “Double win”) into the following ciphertext:

“PNQQ YBT GNWIX BNYM TSJ XYTSJ”

4. Apply El-Kindi attack approach (frequency analysis) to crack the key K that was used to encrypt an english plaintext into the following ciphertext:

“KWJVVZJSHD FSFQDXNX NX F MNXYTWNHFQ YJHMSNVZJ ZXJI YT
GWJFP XNRUQJ HNUMJWX. GD FSFQDENS L YMJ QJYYJW KWJVVZJSHD
INXYWNGZYNTS NS F HNUMJW YJCY, HWDUYTFSFQDXYX HTZQI
NIJSYNKD UFYYJWSX FSI UTYJSYNFQQD LZJXX YMJ TWNLNSFQ
RJXXFLJ.”

5. Apply El-Kindi again to crack the key K that was used to encrypt an arabic plaintext into the following ciphertext (assume 28 letters — from “أ” to “ي”):

“جويسخ جوتفانخ جوكوثج وشهجع جوجعفاكث”

Exercise 3 (Affine Cipher)

1. Consider the Affine cipher with parameters $a = 3$ and $b = 7$ (i.e., $\mathcal{E}(x) = a \cdot x + b [26]$). Encrypt the following plaintext: “ARTIFICIAL INTELLIGENCE”
2. Could you decrypt the following ciphertext knowing that it was claimed to have been encrypted using the Affine cipher with parameters $a = 6$ and $b = 7$?

“CW. GBRULD ZHWLN”

3. Could you decrypt the following ciphertext knowing that it was claimed to have been encrypted using the Affine cipher with parameters $a = 9$ and $b = 5$?

“GC. RFCZJ ABDSZL”

4. The following ciphertext was encrypted using Affine cipher. Perform cryptanalysis over the ciphertext to crack the key and disclose the original plaintext. Information about the used parameters were revealed: $b = 4$ and $11 \leq a \leq 19$?

“MHWZPICHEZTW KY CIID”

5. Perform the same attack against the below ciphertext. The following information about the used parameters were revealed: $b = 25$ and $3 \leq a \leq 9$?

“PTDMP GZNR CT ZYPBODZ”

Exercise 4 (Hill Cipher)

1. Use Hill cipher to encrypt the word “BE CAREFUL” using the following key K :

$$K = \begin{pmatrix} 7 & 5 \\ 7 & 11 \end{pmatrix}$$

2. Use Hill cipher to encrypt the word “ENSIA” using the following key K . Then, use the same key to decrypt the message “LXOQCHNR”:

$$K = \begin{pmatrix} 6 & 7 \\ 9 & 5 \end{pmatrix}$$

3. Use Hill cipher with key K to decrypt the ciphertext “ZDMWCLTV”. Note that a letter “Z” was added as a padding to the original plaintext:

$$K = \begin{pmatrix} 3 & 1 \\ 5 & 14 \end{pmatrix}$$

4. Perform cryptanalysis on Hill cipher that was used to encrypt a plaintext into the ciphertext “NVGLCH”. The key K that was used had the below form, where $b = 4$ and $c = 5$. Also, the parameters $0 < a \leq 7$ and $0 < d \leq 4$, and $(a + d) - 2\sqrt{5 + (b \cdot c)} = 0$.

$$K = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

5. Consider Hill cipher with the below key K . Is this key valid? Justify your answer. In case the key is valid, encrypt the plaintext “IJMM”. Also, decrypt the message “REVE”. Is it enough to just know the tuple (“IJMM”, ciphertext) to crack the key K ? Same question for the tuple (plaintext, “REVE”). Think about a tuple (plaintext, ciphertext) that can be used to run a known-plaintext attack against Hill to crack K .

$$K = \begin{pmatrix} 11 & 8 \\ 3 & 7 \end{pmatrix}$$